

AUDIT SURFACE FOR

*Capability Matters: A Casebook*

# Validation & Audit

INDEXES · PER-CASE REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

EDITION · 15 JUNE 2026

SECTION

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## What this document is for

**Validation and Audit** is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the CLOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

**On verification.** The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

# The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE

Case 5

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS

Case 18

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

|    |                                                                                         |                      |       |
|----|-----------------------------------------------------------------------------------------|----------------------|-------|
| 1  | USS Fitzgerald & USS John S. McCain                                                     | 2017                 | T·K·N |
| 2  | Marine Corps Training in the INDOPACOM AOR                                              | ongoing              | T·K   |
| 3  | F-35 Sustainment & Maintainer Shortage                                                  | ongoing              | T·K·D |
| 4  | Military Fratricide — Desert Storm to Afghanistan                                       | 1991 – 2014          | T·H·K |
| 5  | Operation Eagle Claw                                                                    | 1980                 | T·K   |
| 8  | Patriot Missile / Dhahran                                                               | 1991                 | D·H·K |
| 17 | GAO Weapon–System Sustainment Reviews — Operating Costs without Decision–Grade Data     | 2022 (GAO-22-104533) | G·K   |
| 18 | U.S. Nuclear Navy / Rickover Training Model                                             | 1954 – present       | T·K·N |
| 21 | MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard (series since 1968) | 2020 revision        | G·K   |
| 33 | GIFT and the Adoption Gap                                                               | 2012 – present       | K·G·N |
| 37 | Navy Surface Warfare Readiness Reform                                                   | 2018 – present       | T·K·N |

|     |                                                                       |                |       |
|-----|-----------------------------------------------------------------------|----------------|-------|
| 42  | DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks | 2009 – 2014    | H·K·D |
| 54  | USS Vincennes & Iran Air Flight 655                                   | 1988           | H·T   |
| 64  | Stanislav Petrov / 1983 False Alert                                   | 1983           | H·T   |
| 138 | Mark 14 Torpedo Failures                                              | 1941 – 1943    | T·K·G |
| 141 | V-22 Osprey                                                           | 1991 – present | T·H·N |
| 167 | 9/11 Intelligence Sharing Failures                                    | 1996 – 2001    | G·K   |

## AVIATION

### STANDOUT FAILURE

#### Case 7

**Boeing 737 MAX / MCAS (2018–19).** A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

### STANDOUT SUCCESS

#### Case 70

**Crew Resource Management & CAST (1981–).** First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

|     |                                                              |                |       |
|-----|--------------------------------------------------------------|----------------|-------|
| 6   | AeroPerú Flight 603                                          | 1996           | H·T   |
| 7   | Boeing 737 MAX / MCAS                                        | 2018 – 2019    | D·T·H |
| 12  | Boeing 737 Rudder Hardovers                                  | 1991, 1994     | H·D   |
| 28  | Air France Flight 447                                        | 2009           | T·H   |
| 55  | Kegworth / British Midland 92                                | 1989           | T·H·K |
| 56  | Asiana Airlines Flight 214                                   | 2013           | T·H   |
| 57  | Helios Airways Flight 522                                    | 2005           | T·H   |
| 58  | TransAsia Airways Flight 235                                 | 2015           | T·H   |
| 63  | Eastern Air Lines Flight 401                                 | 1972           | H·T   |
| 68  | Air Canada Chatbot Liability — Delegation Without Revocation | 2022 – 2024    | D·K·N |
| 70  | Crew Resource Management & CAST                              | 1981 – present | T·H·N |
| 72  | Korean Air Safety Transformation                             | 2000 – present | T·N   |
| 82  | Colgan Air Flight 3407                                       | 2009           | T     |
| 83  | Atlas Air Flight 3591                                        | 2019           | T·K   |
| 88  | F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap | 2008 – 2012    | H·K·N |
| 111 | Aviation Safety Reporting System (ASRS)                      | 1976 – present | T·K·N |

|     |                                                                                  |                      |       |
|-----|----------------------------------------------------------------------------------|----------------------|-------|
| 118 | EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation                  | 1996 – 2002          | H·K·G |
| 119 | TCAS — Coordinated Collision Avoidance and the Überlingen Lesson<br>Version 7.1) | 1981 – 2008 (TCAS II | H·K·G |
| 164 | NASA Saturn V Documentation                                                      | 1972 – present       | K     |
| 175 | Singapore Airlines Safety Transformation                                         | 1980s – present      | T·N   |

## HEALTHCARE

STANDOUT  
FAILURE

## Case 61

**EHR / CPOE Implementation (2005–).** Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT  
SUCCESS

## Case 109

**WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

|     |                                                                                  |                |           |
|-----|----------------------------------------------------------------------------------|----------------|-----------|
| 20  | California Nurse Staffing Ratios — Legislating a Capability Requirement          | 2004 – 2010    | G·K       |
| 32  | ACGME 80-Hour Resident Duty-Hour Reform                                          | 2003–2017      | T·K·N     |
| 34  | Implementation Science in Healthcare — The 17-Year Gap                           | ongoing        | K·G·N     |
| 59  | Therac-25                                                                        | 1985 – 1987    | H·D·G     |
| 61  | EHR / CPOE Implementation                                                        | 2005 – present | H·D·G     |
| 71  | Keystone ICU / Pronovost Checklist                                               | 2004 – present | T·N       |
| 74  | TREWS / Sepsis Watch                                                             | 2018 – 2022    | H·K·G     |
| 75  | Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff | 2024 – 2025    | H·K·N     |
| 79  | ChatGPT in Healthcare — Hallucination Cases                                      | 2023 – present | H·D       |
| 86  | Mid Staffordshire NHS Foundation Trust                                           | 2005 – 2009    | G·N·K     |
| 93  | Epic Sepsis Model                                                                | 2017 – 2021    | D·G·N     |
| 102 | Medical Errors as Systemic Failure                                               | 1999 – present | T·H·N·K·G |
| 105 | Vioxx Withdrawal                                                                 | 1999 – 2004    | G·D       |
| 109 | WHO Surgical Safety Checklist                                                    | 2008 – present | T·N       |
| 112 | Bristol Heart Babies Reform                                                      | 1984 – present | G·N       |
| 125 | Annual-Screening UI Redesign + CDS at University of Missouri Health Care         | 2020           | T·D·N     |
| 126 | Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal    | 2019 – 2024    | T·D·N     |

|     |                                                                          |                |       |
|-----|--------------------------------------------------------------------------|----------------|-------|
| 129 | COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case           | 2022 – 2024    | T·K·D |
| 130 | Radiology AI Miscalibration                                              | 2018 – present | H·K·D |
| 132 | AlphaFold — Protein Structure Prediction                                 | 2020 – present | T     |
| 136 | DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback | 2020           | T·K·D |
| 150 | Theranos                                                                 | 2003 – 2018    | G·D   |
| 173 | TeamSTEPPS                                                               | 2006 – present | T·N   |
| 174 | Tylenol Recall                                                           | 1982           | G·N   |

## EDUCATION AT SCALE

### STANDOUT FAILURE

#### Case 149

**Summit Learning / Personalized Learning Rollout (2014–19).** A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

### STANDOUT SUCCESS

#### Case 110

**Georgia State University Predictive Analytics (2012–).** 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

|     |                                                           |                 |             |
|-----|-----------------------------------------------------------|-----------------|-------------|
| 38  | Cognitive Tutor / Carnegie Learning                       | 1990s – present | T           |
| 89  | Tennessee Voluntary Pre-K Study                           | 2009–2018       | G·D         |
| 90  | Algorithmic Bias in Educational Predictive Analytics      | ongoing         | G·H·D       |
| 104 | Atlanta Public Schools Cheating Scandal                   | 2009 – 2015     | G·N         |
| 110 | Georgia State University Predictive Analytics             | 2012 – present  | T·K         |
| 146 | inBloom                                                   | 2014            | G           |
| 149 | Summit Learning / Personalized Learning Rollout           | 2014–2019       | G·T·K       |
| 171 | xAPI / Total Learning Architecture — Interoperability Gap | ongoing         | K·G         |
| 194 | The Discipline We Build Next                              | ongoing         | T·K·N·G·H·D |

## ENERGY

STANDOUT  
FAILURE

## Case 69

**Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT  
SUCCESS

## Case 172

**INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

11 Sago Mine Disaster 2006

N-T-K

60 Northeast Blackout 2003

H-K

69 Three Mile Island 1979

T-H

85 Deepwater Horizon 2010

T-N-K

87 Upper Big Branch Mine Explosion 2010

N-G-K

137 Texas City BP Refinery Explosion 2005

N-T-K-G

142 Davis-Besse Reactor Head Corrosion 2002

N-K-G

143 Fukushima Daiichi 2011

N-G-K

154 Camp Fire / PG&amp;E 2018

G-N-K

172 INPO and the Nuclear Academy 1979 – present

T-K-G

## GOVERNMENT

STANDOUT  
FAILURE

## Case 162

**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.

## NOTE

## Case 111

No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14 Healthcare.gov Launch 2013

K-T-G

101 VA Wait-Time Scandal 2014

G-K-N

103 LIBOR Manipulation 2003 – 2012

G-N

131 Predictive Policing — PredPol 2011 – present

G-H-D

|     |                               |             |       |
|-----|-------------------------------|-------------|-------|
| 166 | Bernard Madoff / SEC Failures | 1992 – 2008 | G-K-N |
|-----|-------------------------------|-------------|-------|

INDUSTRIAL

|                  |          |                                                                                                                                                                                                                                              |
|------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STANDOUT FAILURE | Case 147 | <b>Bhopal (1984).</b> Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility. |
| STANDOUT SUCCESS | Case 73  | <b>Toyota Production System / Andon Cord (1950s–).</b> Any operator can stop the line. The smallest authority–architecture intervention in the dataset, and the most durably copied.                                                         |

|     |                                       |                 |         |
|-----|---------------------------------------|-----------------|---------|
| 30  | Kodak Digital Camera Stagnation       | 1975–2012       | D-K-N   |
| 73  | Toyota Production System / Andon Cord | 1950s – present | N-G     |
| 84  | Takata Airbag Inflators               | 2008 – 2023     | D-G     |
| 92  | Volkswagen Dieselgate                 | 2015            | D-G     |
| 139 | GM Ignition Switch                    | 2002 – 2014     | D-G     |
| 147 | Bhopal                                | 1984            | T-K-N-G |
| 148 | Grenfell Tower                        | 2017            | G-T-K-N |
| 153 | Hyatt Regency Walkway Collapse        | 1981            | D-G     |

TECHNOLOGY

|                  |          |                                                                                                                                                                                                                                                                                                         |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STANDOUT FAILURE | Case 66  | <b>UK Post Office Horizon Scandal (1999–2015).</b> Institutional refusal to believe sub–postmasters’ reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest–running operator–feedback failure in the dataset. |
| NOTE             | Case 132 | The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability–engineering done as a discipline.                                                                                                                   |

|    |                                |                |       |
|----|--------------------------------|----------------|-------|
| 10 | Knight Capital Trading Loss    | 2012           | D-K   |
| 31 | BlackBerry Touchscreen Inertia | 2007–2013      | D-N-K |
| 66 | UK Post Office Horizon Scandal | 1999 – 2015    | G-H-K |
| 80 | AI-Augmented Coding Tools      | 2021 – present | T-H   |
| 91 | Wells Fargo Fake Accounts      | 2011 – 2016    | G-N   |



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|-----|--------------------------------|-------------|-------|
| 144 | CrowdStrike Falcon Outage      | 2024        | D·K·G |
| 145 | TSB Bank IT Migration          | 2018        | H·G   |
| 151 | Cambridge Analytica / Facebook | 2014 – 2018 | G     |
| 152 | Equifax Data Breach            | 2017        | G·K   |

AI IN EDUCATION

|     |                                                                        |      |       |
|-----|------------------------------------------------------------------------|------|-------|
| 135 | LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models | 2025 | T·K·N |
|-----|------------------------------------------------------------------------|------|-------|

K-12 EDUCATION

|    |                                                                                  |              |       |
|----|----------------------------------------------------------------------------------|--------------|-------|
| 36 | Growth-Mindset National Experiment — Heterogeneous Effects                       | 2019         | D·N·K |
| 52 | Chen et al. — Rural China AI Devices and the Equity-Direction Finding            | 2025         | T·K·D |
| 97 | School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism | 2010s – 2022 | G·K·N |

K-12 MATHEMATICS

|     |                                                                       |                |       |
|-----|-----------------------------------------------------------------------|----------------|-------|
| 49  | ASSISTments — National Replication and Long-Term Follow-Through       | 2014 – present | T·K·D |
| 127 | Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive | 2007 – 2014    | T·K·D |

K-12 SCIENCE

|    |                                                             |      |       |
|----|-------------------------------------------------------------|------|-------|
| 51 | Zhang/Scardamalia — Knowledge Building Across Three Cohorts | 2009 | T·K·D |
|----|-------------------------------------------------------------|------|-------|

AIR TRAFFIC MANAGEMENT

|    |                                                                                             |      |       |
|----|---------------------------------------------------------------------------------------------|------|-------|
| 25 | Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change | 2005 | G·K·H |
|----|---------------------------------------------------------------------------------------------|------|-------|

ANESTHESIOLOGY

|     |                                                                      |                |       |
|-----|----------------------------------------------------------------------|----------------|-------|
| 116 | Anesthesia Monitoring Standards and the APSF — The Mortality Decline | 1986 – present | H·K·G |
|-----|----------------------------------------------------------------------|----------------|-------|

AUTONOMOUS SYSTEMS

|                  |          |                                                                                                                                                                                                                                   |
|------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STANDOUT FAILURE | Case 62  | Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles. |
| NOTE             | Case 194 | No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (The Discipline We Build Next) for the open question this volume closes on.  |

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|    |                           |      |         |
|----|---------------------------|------|---------|
| 62 | Uber ATG / Tempe Fatality | 2018 | T·N·G·H |
|----|---------------------------|------|---------|

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|----|----------------------------------------|----------------|---------|
| 78 | Tesla Autopilot — Recurring Fatalities | 2016 – present | T·N·G·H |
|----|----------------------------------------|----------------|---------|

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|-----|-----------------------------------|------|-------|
| 192 | Cruise Robotaxi — Pedestrian Drag | 2023 | G·D·H |
|-----|-----------------------------------|------|-------|

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## AUTONOMOUS VEHICLES

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|     |                                                                         |      |       |
|-----|-------------------------------------------------------------------------|------|-------|
| 156 | Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment | 2023 | G·K·N |
|-----|-------------------------------------------------------------------------|------|-------|

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|-----|-------------------------------------------------------------------------------------|-------------|-------|
| 183 | Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact | 2023 / 2025 | G·K·N |
|-----|-------------------------------------------------------------------------------------|-------------|-------|

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|-----|----------------------------------------------------------------------------------|-------------|-------|
| 184 | CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver | 2020 – 2024 | G·K·D |
|-----|----------------------------------------------------------------------------------|-------------|-------|

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## AVIATION INFRASTRUCTURE

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|    |                                          |             |       |
|----|------------------------------------------|-------------|-------|
| 23 | FAA NextGen and the ADS-B Out Transition | 2003 – 2020 | G·D·K |
|----|------------------------------------------|-------------|-------|

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## AVIATION SAFETY

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|    |                                                                             |              |       |
|----|-----------------------------------------------------------------------------|--------------|-------|
| 22 | FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule | 1988 – 2010s | G·D·K |
|----|-----------------------------------------------------------------------------|--------------|-------|

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|     |                                                                                   |              |       |
|-----|-----------------------------------------------------------------------------------|--------------|-------|
| 180 | FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike | 1992 – 2000s | G·K·N |
|-----|-----------------------------------------------------------------------------------|--------------|-------|

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## CLINICAL CARE

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|    |                                                                 |      |       |
|----|-----------------------------------------------------------------|------|-------|
| 43 | I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer | 2014 | H·K·N |
|----|-----------------------------------------------------------------|------|-------|

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|     |                                                               |      |       |
|-----|---------------------------------------------------------------|------|-------|
| 120 | Bar-Code Medication Administration — Cue at the Point of Care | 2010 | H·K·D |
|-----|---------------------------------------------------------------|------|-------|

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## CLINICAL MEDICINE

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|    |                                                             |      |       |
|----|-------------------------------------------------------------|------|-------|
| 95 | Racial Bias in Pain Assessment — The False-Belief Mechanism | 2016 | T·K·N |
|----|-------------------------------------------------------------|------|-------|

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|     |                                         |      |       |
|-----|-----------------------------------------|------|-------|
| 113 | Removing the Race Coefficient from eGFR | 2021 | D·G·N |
|-----|-----------------------------------------|------|-------|

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## CLINICAL/TRANSLATIONAL RESEARCH

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|     |                                                                 |             |     |
|-----|-----------------------------------------------------------------|-------------|-----|
| 178 | Team Science Training for Clinical and Translational Scientists | 2020 – 2022 | K·N |
|-----|-----------------------------------------------------------------|-------------|-----|

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## CORPORATE L&D

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|    |                                                        |      |     |
|----|--------------------------------------------------------|------|-----|
| 35 | Training Transfer — The Gap Between Learning and Doing | 2010 | K·N |
|----|--------------------------------------------------------|------|-----|

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|----|-------------------------------------------------------------------------------|----------------|-----|
| 46 | High-Impact Learning System — Engineering the Environment, Not Just the Event | 2001 – present | K·N |
|----|-------------------------------------------------------------------------------|----------------|-----|

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|     |                                                               |                |     |
|-----|---------------------------------------------------------------|----------------|-----|
| 108 | The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops | 1959 – present | K·N |
|-----|---------------------------------------------------------------|----------------|-----|

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|-----|----------------------------------------------------------------------|----------------|-----|
| 124 | Brinkerhoff Success Case Method — Tails as the Evaluation Instrument | 2005 – present | K·N |
|-----|----------------------------------------------------------------------|----------------|-----|

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## CRIMINAL JUSTICE

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**99** COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018 **D-K-N**

## DATA GOVERNANCE

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**190** CARE Principles — Indigenous Data Governance as a Replaced Regime 2019 – 2020 (principles); ongoing **G-N**

## DIGITAL IDENTITY

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**187** Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India 2018 – 2025 **G-N-H**

## E-GOVERNMENT

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**27** Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present **G-K-N**

## ED-TECH

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**96** Algorithmic Bias in Automated Exam Proctoring 2022 **D-N-K**

## EDUCATION (NORWAY)

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**182** Norway's National Expert Commission on Learning Analytics 2022 – 2023 **G-K-N**

## EDUCATION AT SCALE

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**40** Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T-D**

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**100** Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2024 **D-K-N**

## FINANCE

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**94** Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2018 – 2022 **D-G-N**

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**107** Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025 **D-G-N**

## FINANCIAL SERVICES

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**163** Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021 **D-K-N**

## GENERAL AVIATION

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**13** Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H-N-K**

## GLOBAL HEALTH

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**47** PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023 **K-N-D**

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**188** Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018 **H-N**

## GOVERNMENT

|     |                                                                                                    |       |
|-----|----------------------------------------------------------------------------------------------------|-------|
| 98  | UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020 | D·K·N |
| 162 | Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict 2016 – 2023        | D·G·N |
| 191 | NYC Local Law 144 — Bias Audits as Governance Artifact 2023 – present                              | D·G·K |

## GOVERNMENT/WELFARE (NETHERLANDS)

|     |                                                                              |       |
|-----|------------------------------------------------------------------------------|-------|
| 155 | Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds 2014 – 2020 | D·G·N |
|-----|------------------------------------------------------------------------------|-------|

## HEALTH PROFESSIONS EDUCATION

|     |                                                              |     |
|-----|--------------------------------------------------------------|-----|
| 117 | Interprofessional Education and the Evidence Gap 2013 – 2015 | K·N |
|-----|--------------------------------------------------------------|-----|

## HEALTHCARE/ONCOLOGY

|    |                                                                           |       |
|----|---------------------------------------------------------------------------|-------|
| 67 | Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018 | D·K·N |
|----|---------------------------------------------------------------------------|-------|

## HIGHER EDUCATION

|     |                                                                                                                    |       |
|-----|--------------------------------------------------------------------------------------------------------------------|-------|
| 128 | OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023                                    | T·K·D |
| 133 | Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction 2024                     | D·K·N |
| 157 | Enrollment-Algorithm Yield Optimization Across U.S. Higher Education (Brookings synthesis 2021)                    | T·K·N |
| 158 | Crisis Point — Merit-Aid Leveraging at Public Flagships 2001 – 2017 (Burd IPEDS analysis); 2024 (Lifting the Veil) | T·K·N |
| 159 | GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022                                                      | D·K·N |
| 160 | USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024                           | D·K·N |
| 161 | In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure 2019 – 2022                 | D·K·N |
| 170 | Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025                                  | T·K·N |

## HIGHER EDUCATION (AFRICA)

|     |                                                                   |       |
|-----|-------------------------------------------------------------------|-------|
| 169 | Data Privacy and Learning Analytics on the African Continent 2022 | K·G·N |
|-----|-------------------------------------------------------------------|-------|

## HIGHER EDUCATION (BRAZIL)

|     |                                                                                    |     |
|-----|------------------------------------------------------------------------------------|-----|
| 123 | MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) 2024 | K·N |
|-----|------------------------------------------------------------------------------------|-----|

## HIGHER EDUCATION (LATIN AMERICA)

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|            |                                                                                    |                |            |
|------------|------------------------------------------------------------------------------------|----------------|------------|
| <b>181</b> | <b>LALA — Building Learning-Analytics Governance Capacity Across Latin America</b> | 2017<br>– 2020 | <b>K·N</b> |
|------------|------------------------------------------------------------------------------------|----------------|------------|

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## HIGHER ED (UK)

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|            |                                                                     |             |              |
|------------|---------------------------------------------------------------------|-------------|--------------|
| <b>115</b> | <b>Open University 'Ethical Use of Student Data' and OU Analyse</b> | 2014 – 2025 | <b>G·K·N</b> |
|------------|---------------------------------------------------------------------|-------------|--------------|

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## HIGHER-ED ANALYTICS

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|            |                                                                      |             |              |
|------------|----------------------------------------------------------------------|-------------|--------------|
| <b>106</b> | <b>Purdue Course Signals — The Reverse-Causality Retention Claim</b> | 2012 – 2013 | <b>D·K·N</b> |
|------------|----------------------------------------------------------------------|-------------|--------------|

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## HOSPITAL PHARMACY

---

|            |                                                              |      |              |
|------------|--------------------------------------------------------------|------|--------------|
| <b>186</b> | <b>Australian Hospital-Pharmacy Technician Role Redesign</b> | 2016 | <b>D·N·H</b> |
|------------|--------------------------------------------------------------|------|--------------|

---

## IMPLEMENTATION SCIENCE

---

|            |                                                                                       |       |            |
|------------|---------------------------------------------------------------------------------------|-------|------------|
| <b>179</b> | <b>Implementation-Science Training — Stated Goals Outrunning Operational Practice</b> | 2020s | <b>K·N</b> |
|------------|---------------------------------------------------------------------------------------|-------|------------|

---

## LEARNING ANALYTICS

---

|           |                                                                                            |             |              |
|-----------|--------------------------------------------------------------------------------------------|-------------|--------------|
| <b>50</b> | <b>The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension</b> | 2016 – 2025 | <b>T·K·D</b> |
|-----------|--------------------------------------------------------------------------------------------|-------------|--------------|

---

|            |                                                                                |             |              |
|------------|--------------------------------------------------------------------------------|-------------|--------------|
| <b>134</b> | <b>Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models</b> | 2021 – 2024 | <b>D·K·N</b> |
|------------|--------------------------------------------------------------------------------|-------------|--------------|

---

|            |                                                                                                |      |            |
|------------|------------------------------------------------------------------------------------------------|------|------------|
| <b>193</b> | <b>Learning Analytics on the African Continent — A Scoping Review as the Present-State Map</b> | 2022 | <b>K·N</b> |
|------------|------------------------------------------------------------------------------------------------|------|------------|

---

## MEDICAL DEVICES

---

|            |                                         |                    |              |
|------------|-----------------------------------------|--------------------|--------------|
| <b>114</b> | <b>Pulse Oximetry Across Skin Tones</b> | 1990 – 2020 – 2025 | <b>D·G·N</b> |
|------------|-----------------------------------------|--------------------|--------------|

---

## MEDICAL EDUCATION

---

|           |                                                                                                  |              |            |
|-----------|--------------------------------------------------------------------------------------------------|--------------|------------|
| <b>41</b> | <b>Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA</b> | 2009 – 2020s | <b>T·K</b> |
|-----------|--------------------------------------------------------------------------------------------------|--------------|------------|

---

|           |                                                  |             |              |
|-----------|--------------------------------------------------|-------------|--------------|
| <b>45</b> | <b>Spaced Education RCTs in Medical Training</b> | 2007 – 2009 | <b>H·K·D</b> |
|-----------|--------------------------------------------------|-------------|--------------|

---

## MEDICAL-DEVICE REGULATION

---

|            |                                                                                           |                |            |
|------------|-------------------------------------------------------------------------------------------|----------------|------------|
| <b>189</b> | <b>Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD</b> | 2014 – present | <b>G·N</b> |
|------------|-------------------------------------------------------------------------------------------|----------------|------------|

---

## MULTIMODAL LEARNING ANALYTICS

---

|           |                                                                                           |      |              |
|-----------|-------------------------------------------------------------------------------------------|------|--------------|
| <b>53</b> | <b>Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account</b> | 2023 | <b>T·K·N</b> |
|-----------|-------------------------------------------------------------------------------------------|------|--------------|

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## NAVAL ENGINEERING

- 
- |           |                                                                         |                |              |
|-----------|-------------------------------------------------------------------------|----------------|--------------|
| <b>19</b> | Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable | 1963 – present | <b>G·K·H</b> |
|-----------|-------------------------------------------------------------------------|----------------|--------------|
- 

## NEUROSCIENCE

- 
- |           |                                                         |             |            |
|-----------|---------------------------------------------------------|-------------|------------|
| <b>16</b> | EU Human Brain Project — Top-Down Vision That Unraveled | 2013 – 2023 | <b>G·N</b> |
|-----------|---------------------------------------------------------|-------------|------------|
- 
- |            |                                                                  |                       |              |
|------------|------------------------------------------------------------------|-----------------------|--------------|
| <b>176</b> | Launching the BRAIN Initiative — Governance of a Grand Challenge | 2011 – 2015 – present | <b>G·K·N</b> |
|------------|------------------------------------------------------------------|-----------------------|--------------|
- 

## NEUROSCIENCE/CONNECTOMICS

- 
- |           |                                                                  |                |              |
|-----------|------------------------------------------------------------------|----------------|--------------|
| <b>81</b> | CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale | 2017 – present | <b>H·K·N</b> |
|-----------|------------------------------------------------------------------|----------------|--------------|
- 

## NUCLEAR ENGINEERING

- 
- |           |                                                                                 |                |              |
|-----------|---------------------------------------------------------------------------------|----------------|--------------|
| <b>76</b> | INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet | 2010 – present | <b>D·H·K</b> |
|-----------|---------------------------------------------------------------------------------|----------------|--------------|
- 

## NUCLEAR POWER

- 
- |           |                                                                             |      |              |
|-----------|-----------------------------------------------------------------------------|------|--------------|
| <b>26</b> | INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding | 2014 | <b>G·K·H</b> |
|-----------|-----------------------------------------------------------------------------|------|--------------|
- 

## NURSING EDUCATION

- 
- |           |                                                                       |      |              |
|-----------|-----------------------------------------------------------------------|------|--------------|
| <b>44</b> | NCSBN National Simulation Study — Licensing the 50% Substitution Rule | 2014 | <b>G·K·D</b> |
|-----------|-----------------------------------------------------------------------|------|--------------|
- 

## PHARMA SAFETY

- 
- |            |                                                                            |             |              |
|------------|----------------------------------------------------------------------------|-------------|--------------|
| <b>168</b> | iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms | 2006 – 2011 | <b>G·D·N</b> |
|------------|----------------------------------------------------------------------------|-------------|--------------|
- 

## RAIL TRANSPORT

- 
- |            |                                                     |                |              |
|------------|-----------------------------------------------------|----------------|--------------|
| <b>177</b> | CIRAS — Confidential Incident Reporting for UK Rail | 1996 – present | <b>G·K·N</b> |
|------------|-----------------------------------------------------|----------------|--------------|
- 

## SOFTWARE ENGINEERING EDUCATION

- 
- |           |                                                                                                        |                 |            |
|-----------|--------------------------------------------------------------------------------------------------------|-----------------|------------|
| <b>48</b> | Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development | 2025 (preprint) | <b>K·N</b> |
|-----------|--------------------------------------------------------------------------------------------------------|-----------------|------------|
- 

## SOFTWARE SUSTAINMENT

- 
- |           |                                                                                   |             |              |
|-----------|-----------------------------------------------------------------------------------|-------------|--------------|
| <b>24</b> | Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed | 1996 – 2000 | <b>G·D·K</b> |
|-----------|-----------------------------------------------------------------------------------|-------------|--------------|
-

SPACE / AEROSPACE

|                  |          |                                                                                                                                                                                                                                                    |
|------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STANDOUT FAILURE | Case 140 | <b>Challenger &amp; Columbia (1986 / 2003).</b> Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine. |
| NOTE             | Case 18  | No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.                                                                                      |

|     |                                                  |             |       |
|-----|--------------------------------------------------|-------------|-------|
| 9   | Mars Climate Orbiter — Unit Mismatch             | 1999        | D-K   |
| 15  | Ariane 5 Flight 501                              | 1996        | D-K-H |
| 65  | NASA EVA-23 — Water Intrusion Inside a Spacesuit | 2013        | H-N-D |
| 140 | Challenger & Columbia                            | 1986 / 2003 | N-K-G |
| 165 | Boeing Starliner                                 | 2019 – 2024 | K-D   |

SURGERY

|     |                                                                         |             |       |
|-----|-------------------------------------------------------------------------|-------------|-------|
| 121 | Surgical Skill Video Peer-Rating Predicts Complications                 | 2013        | H-D-K |
| 122 | Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units | 2009 – 2016 | T-K-H |

TECHNOLOGY

|    |                                                  |             |       |
|----|--------------------------------------------------|-------------|-------|
| 29 | Amazon Hiring AI — Trained Bias, Deprecated 2018 | 2014 – 2018 | D-K-N |
|----|--------------------------------------------------|-------------|-------|

TUTORING

|    |                                                         |      |       |
|----|---------------------------------------------------------|------|-------|
| 77 | Hybrid Human-AI Tutoring — Augmentation, Not Delegation | 2024 | T-K-D |
|----|---------------------------------------------------------|------|-------|

WORKFORCE DEV

|     |                                                                 |                          |       |
|-----|-----------------------------------------------------------------|--------------------------|-------|
| 39  | CIRCUIT — A Scalable, Equity-Centered Research Workforce Model  | 2017 – 2023 (six cycles) | T-K   |
| 185 | Singapore SkillsFuture — National Workforce Capability at Scale | 2015 – present           | G-K-D |

**About this index.** Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

## LENS COURSE INDEX

# The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

## LEN 1

34 cases

---

1 USS Fitzgerald & USS John S. McCain

---

7 Boeing 737 MAX / MCAS

---

14 Healthcare.gov Launch

---

16 EU Human Brain Project — Top-Down Vision That Unraveled

---

17 GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

---

19 Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

---

21 MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

---

22 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

---

23 FAA NextGen and the ADS-B Out Transition

---

24 Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed

---

25 Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

---

26 INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

---

27 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

---

28 Air France Flight 447

---

33 GIFT and the Adoption Gap

---

34 Implementation Science in Healthcare — The 17-Year Gap

---

38 Cognitive Tutor / Carnegie Learning

---

39 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

---

42 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

---

69 Three Mile Island



70 Crew Resource Management & CAST

76 INL / LWRs Control-Room Modernization — Sustainment Research for an Aging Fleet

95 Racial Bias in Pain Assessment — The False-Belief Mechanism

102 Medical Errors as Systemic Failure

110 Georgia State University Predictive Analytics

132 AlphaFold — Protein Structure Prediction

140 Challenger & Columbia

146 inBloom

151 Cambridge Analytica / Facebook

164 NASA Saturn V Documentation

172 INPO and the Nuclear Academy

173 TeamSTEPPS

176 Launching the BRAIN Initiative — Governance of a Grand Challenge

194 The Discipline We Build Next

## LEN 2

53 cases

4 Military Fratricide — Desert Storm to Afghanistan

6 AeroPerú Flight 603

7 Boeing 737 MAX / MCAS

25 Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

28 Air France Flight 447

30 Kodak Digital Camera Stagnation

31 BlackBerry Touchscreen Inertia

35 Training Transfer — The Gap Between Learning and Doing

36 Growth-Mindset National Experiment — Heterogeneous Effects

39 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

40 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

41 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

42 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

43 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

44 NCSBN National Simulation Study — Licensing the 50% Substitution Rule

- 
- 45 Spaced Education RCTs in Medical Training

---

  - 46 High-Impact Learning System — Engineering the Environment, Not Just the Event

---

  - 47 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption

---

  - 48 Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

---

  - 49 ASSISTments — National Replication and Long-Term Follow-Through

---

  - 50 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

---

  - 51 Zhang/Scardamalia — Knowledge Building Across Three Cohorts

---

  - 53 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

---

  - 54 USS Vincennes & Iran Air Flight 655

---

  - 56 Asiana Airlines Flight 214

---

  - 57 Helios Airways Flight 522

---

  - 58 TransAsia Airways Flight 235

---

  - 59 Therac-25

---

  - 60 Northeast Blackout

---

  - 61 EHR / CPOE Implementation

---

  - 62 Uber ATG / Tempe Fatality

---

  - 63 Eastern Air Lines Flight 401

---

  - 64 Stanislav Petrov / 1983 False Alert

---

  - 66 UK Post Office Horizon Scandal

---

  - 70 Crew Resource Management & CAST

---

  - 72 Korean Air Safety Transformation

---

  - 73 Toyota Production System / Andon Cord

---

  - 74 TREWS / Sepsis Watch

---

  - 77 Hybrid Human-AI Tutoring — Augmentation, Not Delegation

---

  - 78 Tesla Autopilot — Recurring Fatalities

---

  - 79 ChatGPT in Healthcare — Hallucination Cases

---

  - 80 AI-Augmented Coding Tools

---

  - 81 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

---

  - 93 Epic Sepsis Model

---

  - 106 Purdue Course Signals — The Reverse-Causality Retention Claim

116 Anesthesia Monitoring Standards and the APSF — The Mortality Decline

120 Bar-Code Medication Administration — Cue at the Point of Care

121 Surgical Skill Video Peer-Rating Predicts Complications

122 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

127 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive

128 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

144 CrowdStrike Falcon Outage

185 Singapore SkillsFuture — National Workforce Capability at Scale

## LEN 3

51 cases

1 USS Fitzgerald & USS John S. McCain

3 F-35 Sustainment & Maintainer Shortage

12 Boeing 737 Rudder Hardovers

18 U.S. Nuclear Navy / Rickover Training Model

26 INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

29 Amazon Hiring AI — Trained Bias, Deprecated 2018

30 Kodak Digital Camera Stagnation

31 BlackBerry Touchscreen Inertia

45 Spaced Education RCTs in Medical Training

49 ASSISTments — National Replication and Long-Term Follow-Through

50 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

51 Zhang/Scardamalia — Knowledge Building Across Three Cohorts

52 Chen et al. — Rural China AI Devices and the Equity-Direction Finding

65 NASA EVA-23 — Water Intrusion Inside a Spacesuit

70 Crew Resource Management & CAST

85 Deepwater Horizon

86 Mid Staffordshire NHS Foundation Trust

88 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

98 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

99 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

100 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

- 
- 112 Bristol Heart Babies Reform
- 
- 118 EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation
- 
- 119 TCAS — Coordinated Collision Avoidance and the Überlingen Lesson
- 
- 125 Annual-Screening UI Redesign + CDS at University of Missouri Health Care
- 
- 126 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal
- 
- 127 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive
- 
- 128 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale
- 
- 129 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case
- 
- 133 Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction
- 
- 134 Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models
- 
- 135 LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models
- 
- 136 DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback
- 
- 137 Texas City BP Refinery Explosion
- 
- 140 Challenger & Columbia
- 
- 141 V-22 Osprey
- 
- 143 Fukushima Daiichi
- 
- 147 Bhopal
- 
- 148 Grenfell Tower
- 
- 157 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education
- 
- 158 Crisis Point — Merit-Aid Leveraging at Public Flagships
- 
- 159 GAO Online Program Manager Oversight Gap (GAO-22-104463)
- 
- 160 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)
- 
- 161 In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure
- 
- 163 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model
- 
- 167 9/11 Intelligence Sharing Failures
- 
- 170 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions
- 
- 172 INPO and the Nuclear Academy
- 
- 173 TeamSTEPPS
- 
- 191 NYC Local Law 144 — Bias Audits as Governance Artifact
- 
- 194 The Discipline We Build Next
-

## LEN 4

65 cases

---

**32** ACGME 80-Hour Resident Duty-Hour Reform

---

**35** Training Transfer — The Gap Between Learning and Doing

---

**37** Navy Surface Warfare Readiness Reform

---

**38** Cognitive Tutor / Carnegie Learning

---

**43** I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

---

**46** High-Impact Learning System — Engineering the Environment, Not Just the Event

---

**47** PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption

---

**70** Crew Resource Management & CAST

---

**71** Keystone ICU / Pronovost Checklist

---

**74** TREWS / Sepsis Watch

---

**75** Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

---

**82** Colgan Air Flight 3407

---

**83** Atlas Air Flight 3591

---

**84** Takata Airbag Inflators

---

**86** Mid Staffordshire NHS Foundation Trust

---

**87** Upper Big Branch Mine Explosion

---

**89** Tennessee Voluntary Pre-K Study

---

**90** Algorithmic Bias in Educational Predictive Analytics

---

**91** Wells Fargo Fake Accounts

---

**92** Volkswagen Dieselgate

---

**93** Epic Sepsis Model

---

**94** Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity

---

**95** Racial Bias in Pain Assessment — The False-Belief Mechanism

---

**97** School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

---

**101** VA Wait-Time Scandal

---

**102** Medical Errors as Systemic Failure

---

**103** LIBOR Manipulation

---

**104** Atlanta Public Schools Cheating Scandal

---

- 
- 105 Vioxx Withdrawal

---

  - 107 Fintech Lending Fairness Audit — When Including Race Reduces Disparity

---

  - 108 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

---

  - 109 WHO Surgical Safety Checklist

---

  - 110 Georgia State University Predictive Analytics

---

  - 111 Aviation Safety Reporting System (ASRS)

---

  - 112 Bristol Heart Babies Reform

---

  - 113 Removing the Race Coefficient from eGFR

---

  - 114 Pulse Oximetry Across Skin Tones

---

  - 115 Open University 'Ethical Use of Student Data' and OU Analyse

---

  - 117 Interprofessional Education and the Evidence Gap

---

  - 123 MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

---

  - 124 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

---

  - 125 Annual-Screening UI Redesign + CDS at University of Missouri Health Care

---

  - 126 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

---

  - 130 Radiology AI Miscalibration

---

  - 137 Texas City BP Refinery Explosion

---

  - 139 GM Ignition Switch

---

  - 150 Theranos

---

  - 155 Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds

---

  - 156 Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment

---

  - 166 Bernard Madoff / SEC Failures

---

  - 169 Data Privacy and Learning Analytics on the African Continent

---

  - 171 xAPI / Total Learning Architecture — Interoperability Gap

---

  - 177 CIRAS — Confidential Incident Reporting for UK Rail

---

  - 178 Team Science Training for Clinical and Translational Scientists

---

  - 179 Implementation-Science Training — Stated Goals Outrunning Operational Practice

---

  - 180 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

---

  - 181 LALA — Building Learning-Analytics Governance Capacity Across Latin America

---

  - 182 Norway's National Expert Commission on Learning Analytics

---

|     |                                                                                         |
|-----|-----------------------------------------------------------------------------------------|
| 183 | Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact     |
| 184 | CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver        |
| 185 | Singapore SkillsFuture — National Workforce Capability at Scale                         |
| 186 | Australian Hospital-Pharmacy Technician Role Redesign                                   |
| 188 | Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface     |
| 190 | CARE Principles — Indigenous Data Governance as a Replaced Regime                       |
| 193 | Learning Analytics on the African Continent — A Scoping Review as the Present-State Map |

## LEN 5

85 cases

|    |                                                                                           |
|----|-------------------------------------------------------------------------------------------|
| 1  | USS Fitzgerald & USS John S. McCain                                                       |
| 2  | Marine Corps Training in the INDOPACOM AOR                                                |
| 3  | F-35 Sustainment & Maintainer Shortage                                                    |
| 4  | Military Fratricide — Desert Storm to Afghanistan                                         |
| 5  | Operation Eagle Claw                                                                      |
| 6  | AeroPerú Flight 603                                                                       |
| 7  | Boeing 737 MAX / MCAS                                                                     |
| 8  | Patriot Missile / Dhahran                                                                 |
| 9  | Mars Climate Orbiter — Unit Mismatch                                                      |
| 10 | Knight Capital Trading Loss                                                               |
| 11 | Sago Mine Disaster                                                                        |
| 12 | Boeing 737 Rudder Hardovers                                                               |
| 13 | Glass-Cockpit Transition in Light Aircraft — Technology Outran Training                   |
| 14 | Healthcare.gov Launch                                                                     |
| 15 | Ariane 5 Flight 501                                                                       |
| 17 | GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data       |
| 18 | U.S. Nuclear Navy / Rickover Training Model                                               |
| 19 | Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable                   |
| 20 | California Nurse Staffing Ratios — Legislating a Capability Requirement                   |
| 21 | MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard                       |
| 27 | Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) |
| 28 | Air France Flight 447                                                                     |

- 
- 29 Amazon Hiring AI — Trained Bias, Deprecated 2018

---

  - 32 ACGME 80-Hour Resident Duty-Hour Reform

---

  - 36 Growth-Mindset National Experiment — Heterogeneous Effects

---

  - 37 Navy Surface Warfare Readiness Reform

---

  - 39 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

---

  - 40 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

---

  - 41 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

---

  - 42 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

---

  - 44 NCSBN National Simulation Study — Licensing the 50% Substitution Rule

---

  - 45 Spaced Education RCTs in Medical Training

---

  - 53 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account

---

  - 54 USS Vincennes & Iran Air Flight 655

---

  - 55 Kegworth / British Midland 92

---

  - 56 Asiana Airlines Flight 214

---

  - 57 Helios Airways Flight 522

---

  - 58 TransAsia Airways Flight 235

---

  - 59 Therac-25

---

  - 63 Eastern Air Lines Flight 401

---

  - 65 NASA EVA-23 — Water Intrusion Inside a Spacesuit

---

  - 67 Watson for Oncology — Delegation Marketed Ahead of Capability

---

  - 68 Air Canada Chatbot Liability — Delegation Without Revocation

---

  - 69 Three Mile Island

---

  - 71 Keystone ICU / Pronovost Checklist

---

  - 75 Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

---

  - 77 Hybrid Human-AI Tutoring — Augmentation, Not Delegation

---

  - 81 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

---

  - 82 Colgan Air Flight 3407

---

  - 85 Deepwater Horizon

---

  - 88 F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap

---

  - 96 Algorithmic Bias in Automated Exam Proctoring

---



**98** UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

**99** COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

**100** Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

**106** Purdue Course Signals — The Reverse-Causality Retention Claim

**116** Anesthesia Monitoring Standards and the APSF — The Mortality Decline

**118** EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation

**119** TCAS — Coordinated Collision Avoidance and the Überlingen Lesson

**120** Bar-Code Medication Administration — Cue at the Point of Care

**121** Surgical Skill Video Peer-Rating Predicts Complications

**122** Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

**129** COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

**135** LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

**136** DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

**141** V-22 Osprey

**144** CrowdStrike Falcon Outage

**147** Bhopal

**152** Equifax Data Breach

**153** Hyatt Regency Walkway Collapse

**157** Enrollment-Algorithm Yield Optimization Across U.S. Higher Education

**158** Crisis Point — Merit-Aid Leveraging at Public Flagships

**159** GAO Online Program Manager Oversight Gap (GAO-22-104463)

**160** USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

**161** In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure

**162** Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict

**163** Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model

**165** Boeing Starliner

**168** iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

**170** Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

**186** Australian Hospital-Pharmacy Technician Role Redesign

**187** Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India

**189** Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

**190** CARE Principles — Indigenous Data Governance as a Replaced Regime

**191** NYC Local Law 144 — Bias Audits as Governance Artifact

## LEN 6

17 cases

**8** Patriot Missile / Dhahran

**14** Healthcare.gov Launch

**25** Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change

**26** INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

**27** Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox)

**33** GIFT and the Adoption Gap

**34** Implementation Science in Healthcare — The 17-Year Gap

**78** Tesla Autopilot — Recurring Fatalities

**102** Medical Errors as Systemic Failure

**133** Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

**134** Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

**139** GM Ignition Switch

**146** inBloom

**151** Cambridge Analytica / Facebook

**165** Boeing Starliner

**171** xAPI / Total Learning Architecture — Interoperability Gap

**174** Tylenol Recall

## LEN 7

106 cases

**12** Boeing 737 Rudder Hardovers

**13** Glass-Cockpit Transition in Light Aircraft — Technology Outran Training

**14** Healthcare.gov Launch

**16** EU Human Brain Project — Top-Down Vision That Unraveled

**17** GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data

**19** Navy SUBSAFE — Requirements as a Non-Negotiable Sustainment Deliverable

**20** California Nurse Staffing Ratios — Legislating a Capability Requirement

**21** MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

- 
- 22 FAA Aging–Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule

---

  - 23 FAA NextGen and the ADS–B Out Transition

---

  - 24 Y2K Remediation — The Aging–System Transition That Worked Because It Was Believed

---

  - 35 Training Transfer — The Gap Between Learning and Doing

---

  - 41 Barsuk SBML — Simulation–Based Mastery Learning Dissemination from Northwestern to the VA

---

  - 43 I–PASS Handoff Bundle — Structuring the Human–to–Human Transfer

---

  - 44 NCSBN National Simulation Study — Licensing the 50% Substitution Rule

---

  - 46 High–Impact Learning System — Engineering the Environment, Not Just the Event

---

  - 47 PEPFAR HIV Training Across 16 Sub–Saharan African Countries — Modality Comparison Under Disruption

---

  - 48 Reflective Practice on a Work–Based Software Engineering Program — Longitudinal Capability Development

---

  - 49 ASSISTments — National Replication and Long–Term Follow–Through

---

  - 50 The Doer Effect at Scale — Replication, AI–Generated Questions, Non–WEIRD Extension

---

  - 51 Zhang/Scardamalia — Knowledge Building Across Three Cohorts

---

  - 52 Chen et al. — Rural China AI Devices and the Equity–Direction Finding

---

  - 53 Multimodal Learning Analytics In–the–Wild — A First–Person Lessons–Learned Account

---

  - 59 Therac–25

---

  - 61 EHR / CPOE Implementation

---

  - 65 NASA EVA–23 — Water Intrusion Inside a Spacesuit

---

  - 66 UK Post Office Horizon Scandal

---

  - 67 Watson for Oncology — Delegation Marketed Ahead of Capability

---

  - 74 TREWS / Sepsis Watch

---

  - 75 Sterile–Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

---

  - 76 INL / LWRS Control–Room Modernization — Sustainment Research for an Aging Fleet

---

  - 77 Hybrid Human–AI Tutoring — Augmentation, Not Delegation

---

  - 78 Tesla Autopilot — Recurring Fatalities

---

  - 79 ChatGPT in Healthcare — Hallucination Cases

---

  - 81 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale

---

  - 84 Takata Airbag Inflators

---

  - 86 Mid Staffordshire NHS Foundation Trust

---

  - 87 Upper Big Branch Mine Explosion

---

- |     |                                                                                  |
|-----|----------------------------------------------------------------------------------|
| 88  | F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap                     |
| 89  | Tennessee Voluntary Pre-K Study                                                  |
| 90  | Algorithmic Bias in Educational Predictive Analytics                             |
| 91  | Wells Fargo Fake Accounts                                                        |
| 92  | Volkswagen Dieselgate                                                            |
| 93  | Epic Sepsis Model                                                                |
| 94  | Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity   |
| 95  | Racial Bias in Pain Assessment — The False-Belief Mechanism                      |
| 97  | School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism |
| 101 | VA Wait-Time Scandal                                                             |
| 103 | LIBOR Manipulation                                                               |
| 104 | Atlanta Public Schools Cheating Scandal                                          |
| 105 | Vioxx Withdrawal                                                                 |
| 107 | Fintech Lending Fairness Audit — When Including Race Reduces Disparity           |
| 108 | The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops                    |
| 110 | Georgia State University Predictive Analytics                                    |
| 112 | Bristol Heart Babies Reform                                                      |
| 113 | Removing the Race Coefficient from eGFR                                          |
| 114 | Pulse Oximetry Across Skin Tones                                                 |
| 115 | Open University 'Ethical Use of Student Data' and OU Analyse                     |
| 116 | Anesthesia Monitoring Standards and the APSF — The Mortality Decline             |
| 117 | Interprofessional Education and the Evidence Gap                                 |
| 118 | EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation                  |
| 119 | TCAS — Coordinated Collision Avoidance and the Überlingen Lesson                 |
| 120 | Bar-Code Medication Administration — Cue at the Point of Care                    |
| 121 | Surgical Skill Video Peer-Rating Predicts Complications                          |
| 122 | Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units          |
| 123 | MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)    |
| 124 | Brinkerhoff Success Case Method — Tails as the Evaluation Instrument             |
| 127 | Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive            |

- 
- 128 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale

---

  - 130 Radiology AI Miscalibration

---

  - 131 Predictive Policing — PredPol

---

  - 132 AlphaFold — Protein Structure Prediction

---

  - 135 LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models

---

  - 137 Texas City BP Refinery Explosion

---

  - 138 Mark 14 Torpedo Failures

---

  - 139 GM Ignition Switch

---

  - 140 Challenger & Columbia

---

  - 142 Davis-Besse Reactor Head Corrosion

---

  - 143 Fukushima Daiichi

---

  - 145 TSB Bank IT Migration

---

  - 146 inBloom

---

  - 147 Bhopal

---

  - 148 Grenfell Tower

---

  - 149 Summit Learning / Personalized Learning Rollout

---

  - 150 Theranos

---

  - 151 Cambridge Analytica / Facebook

---

  - 152 Equifax Data Breach

---

  - 153 Hyatt Regency Walkway Collapse

---

  - 154 Camp Fire / PG&E

---

  - 155 Dutch SyRI — Welfare–Fraud Risk Scoring Halted on Rights Grounds

---

  - 166 Bernard Madoff / SEC Failures

---

  - 167 9/11 Intelligence Sharing Failures

---

  - 168 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

---

  - 169 Data Privacy and Learning Analytics on the African Continent

---

  - 174 Tylenol Recall

---

  - 176 Launching the BRAIN Initiative — Governance of a Grand Challenge

---

  - 177 CIRAS — Confidential Incident Reporting for UK Rail

---

  - 178 Team Science Training for Clinical and Translational Scientists

---

- 179 Implementation–Science Training — Stated Goals Outrunning Operational Practice
- 180 FSF CFIT and ALAR Task Forces — Industry–Level Institution Building After a Spike
- 181 LALA — Building Learning–Analytics Governance Capacity Across Latin America
- 182 Norway's National Expert Commission on Learning Analytics
- 188 Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface
- 189 Japan PMDA — Pre–Approved Change Management as Regulatory Architecture for AI/SaMD
- 192 Cruise Robotaxi — Pedestrian Drag
- 193 Learning Analytics on the African Continent — A Scoping Review as the Present–State Map

## LEN 8

92 cases

- 1 USS Fitzgerald & USS John S. McCain
- 2 Marine Corps Training in the INDOPACOM AOR
- 3 F–35 Sustainment & Maintainer Shortage
- 4 Military Fratricide — Desert Storm to Afghanistan
- 5 Operation Eagle Claw
- 8 Patriot Missile / Dhahran
- 9 Mars Climate Orbiter — Unit Mismatch
- 11 Sago Mine Disaster
- 15 Ariane 5 Flight 501
- 16 EU Human Brain Project — Top–Down Vision That Unraveled
- 18 U.S. Nuclear Navy / Rickover Training Model
- 20 California Nurse Staffing Ratios — Legislating a Capability Requirement
- 22 FAA Aging–Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule
- 23 FAA NextGen and the ADS–B Out Transition
- 24 Y2K Remediation — The Aging–System Transition That Worked Because It Was Believed
- 29 Amazon Hiring AI — Trained Bias, Deprecated 2018
- 32 ACGME 80–Hour Resident Duty–Hour Reform
- 33 GIFT and the Adoption Gap
- 34 Implementation Science in Healthcare — The 17–Year Gap
- 36 Growth–Mindset National Experiment — Heterogeneous Effects
- 40 Duolingo Half–Life Regression — Spaced Repetition at Consumer Scale

- 
- 48** Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development

---

  - 52** Chen et al. — Rural China AI Devices and the Equity-Direction Finding

---

  - 55** Kegworth / British Midland 92

---

  - 60** Northeast Blackout

---

  - 68** Air Canada Chatbot Liability — Delegation Without Revocation

---

  - 72** Korean Air Safety Transformation

---

  - 73** Toyota Production System / Andon Cord

---

  - 76** INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

---

  - 80** AI-Augmented Coding Tools

---

  - 82** Colgan Air Flight 3407

---

  - 83** Atlas Air Flight 3591

---

  - 85** Deepwater Horizon

---

  - 96** Algorithmic Bias in Automated Exam Proctoring

---

  - 97** School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

---

  - 98** UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

---

  - 99** COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate

---

  - 100** Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

---

  - 101** VA Wait-Time Scandal

---

  - 106** Purdue Course Signals — The Reverse-Causality Retention Claim

---

  - 108** The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops

---

  - 111** Aviation Safety Reporting System (ASRS)

---

  - 115** Open University 'Ethical Use of Student Data' and OU Analyse

---

  - 117** Interprofessional Education and the Evidence Gap

---

  - 123** MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

---

  - 124** Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

---

  - 125** Annual-Screening UI Redesign + CDS at University of Missouri Health Care

---

  - 126** Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

---

  - 138** Mark 14 Torpedo Failures

---

  - 139** GM Ignition Switch

---

  - 140** Challenger & Columbia

---

- 
- 141 V-22 Osprey
- 
- 142 Davis-Besse Reactor Head Corrosion
- 
- 143 Fukushima Daiichi
- 
- 145 TSB Bank IT Migration
- 
- 148 Grenfell Tower
- 
- 149 Summit Learning / Personalized Learning Rollout
- 
- 154 Camp Fire / PG&E
- 
- 156 Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment
- 
- 157 Enrollment-Algorithm Yield Optimization Across U.S. Higher Education
- 
- 158 Crisis Point — Merit-Aid Leveraging at Public Flagships
- 
- 159 GAO Online Program Manager Oversight Gap (GAO-22-104463)
- 
- 160 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)
- 
- 161 In re 2U, Inc. Securities Class Action — When Yield Management Crashes Into Disclosure
- 
- 162 Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict
- 
- 163 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model
- 
- 164 NASA Saturn V Documentation
- 
- 165 Boeing Starliner
- 
- 167 9/11 Intelligence Sharing Failures
- 
- 170 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions
- 
- 171 xAPI / Total Learning Architecture — Interoperability Gap
- 
- 172 INPO and the Nuclear Academy
- 
- 173 TeamSTEPPS
- 
- 175 Singapore Airlines Safety Transformation
- 
- 176 Launching the BRAIN Initiative — Governance of a Grand Challenge
- 
- 177 CIRAS — Confidential Incident Reporting for UK Rail
- 
- 178 Team Science Training for Clinical and Translational Scientists
- 
- 179 Implementation-Science Training — Stated Goals Outrunning Operational Practice
- 
- 180 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike
- 
- 181 LALA — Building Learning-Analytics Governance Capacity Across Latin America
- 
- 182 Norway's National Expert Commission on Learning Analytics
-



|     |                                                                                             |
|-----|---------------------------------------------------------------------------------------------|
| 183 | Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact         |
| 184 | CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver            |
| 185 | Singapore SkillsFuture — National Workforce Capability at Scale                             |
| 186 | Australian Hospital-Pharmacy Technician Role Redesign                                       |
| 187 | Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India |
| 188 | Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface         |
| 189 | Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD          |
| 190 | CARE Principles — Indigenous Data Governance as a Replaced Regime                           |
| 191 | NYC Local Law 144 — Bias Audits as Governance Artifact                                      |
| 193 | Learning Analytics on the African Continent — A Scoping Review as the Present-State Map     |
| 194 | The Discipline We Build Next                                                                |

## LEN 9

28 cases

|     |                                                                                           |
|-----|-------------------------------------------------------------------------------------------|
| 8   | Patriot Missile / Dhahran                                                                 |
| 10  | Knight Capital Trading Loss                                                               |
| 13  | Glass-Cockpit Transition in Light Aircraft — Technology Outran Training                   |
| 38  | Cognitive Tutor / Carnegie Learning                                                       |
| 61  | EHR / CPOE Implementation                                                                 |
| 67  | Watson for Oncology — Delegation Marketed Ahead of Capability                             |
| 68  | Air Canada Chatbot Liability — Delegation Without Revocation                              |
| 90  | Algorithmic Bias in Educational Predictive Analytics                                      |
| 94  | Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity            |
| 96  | Algorithmic Bias in Automated Exam Proctoring                                             |
| 107 | Fintech Lending Fairness Audit — When Including Race Reduces Disparity                    |
| 113 | Removing the Race Coefficient from eGFR                                                   |
| 114 | Pulse Oximetry Across Skin Tones                                                          |
| 129 | COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case                            |
| 130 | Radiology AI Miscalibration                                                               |
| 131 | Predictive Policing — PredPol                                                             |
| 132 | AlphaFold — Protein Structure Prediction                                                  |
| 133 | Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction |

|     |                                                                                             |
|-----|---------------------------------------------------------------------------------------------|
| 134 | Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models                     |
| 136 | DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback                    |
| 155 | Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds                            |
| 156 | Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment                     |
| 162 | Australia Robodebt — Algorithmic Debt-Recovery and the Royal Commission Verdict             |
| 168 | iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms                  |
| 169 | Data Privacy and Learning Analytics on the African Continent                                |
| 183 | Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact         |
| 184 | CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver            |
| 187 | Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India |

## LEN 10

19 cases

|     |                                                        |
|-----|--------------------------------------------------------|
| 32  | ACGME 80-Hour Resident Duty-Hour Reform                |
| 33  | GIFT and the Adoption Gap                              |
| 34  | Implementation Science in Healthcare — The 17-Year Gap |
| 37  | Navy Surface Warfare Readiness Reform                  |
| 71  | Keystone ICU / Pronovost Checklist                     |
| 73  | Toyota Production System / Andon Cord                  |
| 79  | ChatGPT in Healthcare — Hallucination Cases            |
| 80  | AI-Augmented Coding Tools                              |
| 89  | Tennessee Voluntary Pre-K Study                        |
| 102 | Medical Errors as Systemic Failure                     |
| 109 | WHO Surgical Safety Checklist                          |
| 138 | Mark 14 Torpedo Failures                               |
| 146 | inBloom                                                |
| 148 | Grenfell Tower                                         |
| 149 | Summit Learning / Personalized Learning Rollout        |
| 173 | TeamSTEPPS                                             |
| 174 | Tylenol Recall                                         |
| 192 | Cruise Robotaxi — Pedestrian Drag                      |
| 194 | The Discipline We Build Next                           |

# References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in casebook/verification-log.md. Future editions will fill in remaining locators as the per-case review closes them out.

**Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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2011

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